

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: ELECTRICAL ENGINEERING DEPARTMENT

2. Subject Name: INSTRUMENTATION DEVICES-II L-T-P: 3-1-0 Credits:4

3. Pre-requisites: Instrumentation Devices-I

4. Syllabus and reference books:

Syllabus: Introducing sensors for force, pressure and acceleration measurement. The design of the signal acquisition circuit for different types of sensors and its applications. Use of feedback principle in Instrumentation and its applications. Principle of operation of fiber optic sensors and its applications. Understanding the principle of Hydraulic and Pneumatic sensors. Introducing different scheme for Signal transmission

Reference Books:

1. Measurement Systems: Application and Design, Ernest O. Doebelin, McGraw Hill Education
2. Principles of Measurement Systems, John P. Bentley, Prentice Hall
3. Instrument Transducers: An Introduction to their performance and design: Hermann K P Neubert. Oxford University Press

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Piezoelectric transducers: construction, properties and equivalent circuit, Charge amplifier, Measurement of force and vibration	4
2.	Tutorial numerical problem on these topics	2
3.	Seismic transducers: Construction and frequency response. Accelerometers: Piezoelectric, Electrodynamical and Capacitive type, MEMS capacitive accelerometer.	4
4.	Tutorial on design issues	2
5.	Ultrasonic transducers: Electromechanical equivalent circuit and transfer function of a piezoelectric transmitter, natural frequencies, crystal oscillator	4
6.	Tutorial numerical problem on these topics	2
7.	Theory of propagation of sound in a medium; Reflection, Refraction and Characteristics Impedance; Ultrasonic Nondestructive Testing and Imaging techniques.	4
8.	Tutorial numerical problem on these topics	2
9.	Optical sources and detection schemes, Fiber optic sensors. Digital transducers: position and speed sensing	4
10.	Tutorial on design issue on the topic	2
11.	Use of feedback principle in Instrumentation and its applications; Elements of pneumatic control systems:	6

	Flapper-nozzle amplifier, Air relay, Valve positioner, Electropneumatic converter. Control Valves: construction and characteristics.	
	Tutorial design issue on these topics	2
	Hydraulic servomotor. A.C servomotor. Stepper motors: constructions, operations and driving schemes. Introduction to Programmable Logic	4
	Tutorial on Design and application	2
	Sources of noise and interference and their reduction techniques, Grounding and Shielding. Signal transmission: 4-20 mA current loops, Serial data communication, Optical fiber links, Wireless sensors.	4
	Tutorial on Numerical problem of the topic	2
Total number of hours		50